

Management of Hemorrhagic Stroke

Introduction

This document provides guidance for the management of hemorrhagic stroke, including blood pressure (BP) control, management of cerebral edema, anticoagulation reversal, and the treatment of hemorrhagic conversion following tPA administration. This guidance is based on current literature and expert consensus.

Blood Pressure (BP) Goals

- For patients with spontaneous intracerebral hemorrhage (ICH) presenting with systolic BP 150–220 mmHg and without contraindication to acute BP treatment, acute lowering of SBP to 140 mmHg is safe and can be effective for improving functional outcomes.
- Target BP <160/90 mmHg in the acute phase to minimize risk of hematoma expansion. Avoid hypotension.

Cerebral Edema Management

- Elevate the head of the bed to 30 degrees to promote venous drainage.
- Consider osmotherapy for increased intracranial pressure (ICP):
 - * Mannitol 20% or 25%: 0.25–1 g/kg IV bolus every 6–8 hours as needed (In line filter required)
 - * Hypertonic saline to maintain serum sodium 145–155 mEq/L
 - 3%: 50 mL over 10-30 minutes or continuous infusion 75 ml/hour
 - 23%: 30 ml bolus over 10 minutes (central line access preferred)
- Monitor for herniation signs and consult neurosurgery for decompressive options.

Anticoagulation Reversal

- Warfarin: Administer IV Vitamin K 10 mg + 4-factor PCC (e.g., Kcentra) based on INR
- DOACs:
 - * Dabigatran: Idarucizumab (Praxbind) 5 g IV
 - * Factor Xa inhibitors (apixaban, rivaroxaban): Andexanet alfa if available or 4F-PCC 50 units/kg or fixed dose strategy (institution specific)
- Heparin: Protamine sulfate 1 mg per 100 units of heparin given in the past 2–3 hours

Post-tPA Hemorrhagic Conversion

- Discontinue any remaining tPA infusion immediately
- Obtain emergent non-contrast CT to confirm hemorrhage

- Administer cryoprecipitate (10 units) to replenish fibrinogen and tranexamic acid 1 g IV over 10 minutes
- May consider aminocaproic acid infusion
- Consider platelet transfusion if antiplatelet therapy was on board
- Neurosurgical consultation and intensive monitoring

References

1. Hemphill JC et al. Guidelines for the Management of Spontaneous Intracerebral Hemorrhage. *Stroke*. 2015;46(7):2032-60. doi:10.1161/STR.0000000000000069
2. Powers WJ et al. Guidelines for the Early Management of Patients With Acute Ischemic Stroke. *Stroke*. 2019;50:e344–e418. doi:10.1161/STR.0000000000000211
3. Frontera JA et al. Guidelines for Reversal of Antithrombotics in Intracranial Hemorrhage. *Neurocrit Care*. 2016;24(1):6–46. doi:10.1007/s12028-015-0222-x